

TAMAL DAS

+353 (085) 778 9972 | tamaldas.personal@gmail.com | github.com/tfortamal | linkedin.com/in/tfortamal | Visa: Stamp 1G

SUMMARY

M.Sc. Data Analytics graduate (First Class Honours, Dublin Business School; B.Sc. Computer Science, First Class Honours) with production experience building and deploying computer vision systems for industrial clients. Skilled in end-to-end model development: dataset curation (50,000+ images), training and validation (YOLOv8, PyTorch), and production optimization (ONNX, C++, quantization). Shipped real-time detection pipelines to Linux servers and validated under live production conditions. Strong foundation in Python and C++, with demonstrated expertise in inference optimization and edge deployment. Based in Dublin on Stamp 1G.

SKILLS

Programming Languages: Python, C++, SQL

Computer Vision & ML: YOLOv8, PyTorch, OpenCV, EasyOCR, COCO/YOLO formats, Anomaly detection, Image classification

Model Optimization: ONNX, Quantization, Non-Maximum Suppression

Deployment & Infrastructure: Linux, Edge Devices

Visualisation: Matplotlib, Power BI, Tableau

EDUCATION

Dublin Business School

Dublin, Ireland

M.Sc. Data Analytics - First Class Honours

June 2026

University of Calcutta

Calcutta, India

B.Sc. Computer Science - First Class Honours (Data Structures, Algorithms, Object-Oriented Design)

2020–2023

PROJECTS

PCB Defect Detection System

Personal Project

YOLOv8n, PyTorch, Python, OpenCV, ONNX, C++

July 2026

- Built an automated PCB defect detection system with YOLOv8n, training incrementally across six balanced data subsets rather than the full 10,628-image dataset at once, mirroring how production models are retrained as new data arrives over time, while keeping a held-out test set fully separate throughout to prevent data leakage.
- Fixed a critical COCO-to-YOLO class-ID indexing bug during dataset conversion that was silently corrupting training labels, then achieved 0.9338 mAP50 on a completely unseen 2,127-image test set, with per-class AP50 ranging from 0.89 to 0.98 across six defect types (missing holes, mouse bites, open circuits, shorts, spurs, spurious copper).
- Optimized the trained model for production deployment: converted YOLOv8 model to ONNX format and applied INT8 quantization, reducing model size by 74% (45MB to 11.5MB) while maintaining 91.1% mAP50 accuracy. Implemented C++ inference pipeline using ONNX Runtime, achieving 12ms per-image latency on CPU (4.1x faster than Python baseline), with Non-Maximum Suppression post-processing to eliminate duplicate detections (99 raw predictions to 2 verified boxes at IOU=0.45).
- Validated optimized model on full 2,127-image test set; documented inference-accuracy trade-offs and demonstrated production-ready optimization mindset for edge deployment scenarios.

Conveyor Belt Material Classification & Foreign Object Detection

R&D Project - DivsoftSolutions

Python, YOLOv8, OpenCV, Image Processing

2023

- Built a real-time industrial inspection system for a steel plant conveyor belt: collected and annotated a 50,000-image dataset on-site, then trained two parallel YOLOv8 models, one for classifying material type across five classes (dolomite, limestone, iron ore fines, pallet, empty belt), one for detecting foreign objects such as metal rods and tin pieces, deliberately separated to prevent the two objectives interfering with each other's learning.
- Deployed the system with real-time alerts streamed to the client's monitoring screen; validated on-site under live production conditions, directly reducing the risk of equipment damage and unplanned downtime, an automated defect-detection workflow closely mirroring industrial optical inspection systems.

Train Wagon Number Detection

R&D Project - DivsoftSolutions

Python, YOLOv8, OpenCV, EasyOCR

2024

- Built an end-to-end real-time detection and OCR pipeline to read 11-digit Indian Railway wagon numbers from live camera feeds at a steel plant rail yard: trained a custom YOLOv8 model on a self-collected on-site dataset to localise number regions under variable lighting and camera angle, then extracted digits with EasyOCR.
- Wrote a check-digit verification script to filter false OCR readings against the Indian Railway numbering scheme, and delivered a live proof-of-concept that replaced a manual logging process, directly reducing mis-recorded wagon movements in the client's production workflow.

EXPERIENCE

R&D Software Developer

DivsoftSolutions India Pvt. Ltd.
2023–2024

- Deployed production YOLOv8 models for real-time material classification on conveyor belt, identifying 5 material types and detecting foreign objects from live camera feeds.
- Built end-to-end OCR pipeline for wagon number identification in steel plant rail yard, reducing manual logging errors and improving production workflow accuracy.
- Collected, annotated, and curated 50,000+ images on-site; designed annotation protocols and implemented quality control for dataset consistency.
- Validated models under live production conditions; shipped to Linux servers with real-time monitoring integration.
- Technologies: Python, YOLOv8, OpenCV, EasyOCR, PyTorch, CNN, Image Processing, Linux

LANGUAGES

English, Bengali, Hindi.

References available upon request